Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Bebout, Lee Ph.D.		
Project Title:	MATS: Microbial Assay Technologies for Space		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/12/2001	**End Date:**	05/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heyenga, Gerard (University of Colorado) Moisander , Pia (NRC NASA Ames and UC Santa Cruz) Bebout, Brad (NASA Ames Research Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Among NASA's primary goals are the imperatives to further our knowledge of how life from Earth can adapt to space conditions. A compelling reason for focusing on complex microbial ecosystems is to understand microbial interactions in response to experimental manipulations or perturbations (such as flight, or introduction of hazardous substances) in their natural context of complex mixed assemblages, rather than in single organism monocultures. To develop a core capability to produce and manipulate small scale microbial communities as defined test systems, and to implement micro-scale technologies for tracking community, process rate, and end-product changes in these test systems as a function of experimental manipulations. The Microbial Assay Technology System (MATS) chamber system will provide the platform for providing research control of key environmental variables. Additionally it will provide the means for detecting changes in several key parameters that microbial communities may alter in responses to external environmental changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The MATS system can be used for a variety of terrestrial benefits including: assessment of biohazardous materials affects on complex microbial systems in small contained systems, development of long-term mixed community biosensors, and optimization of non-fossil fuel gas byproducts from microbial ecosystems.
Task Progress:	Artificial MAT systems are now being constructed for experimentation with parallel use of gene microarrays for nitrogen fixation (nif genes) via collaboration with Dr. Pia Moisander of UC Santa Cruz.
Bibliography Type:	Description: (Last Updated: 02/27/2004)